

Average reward criterion

Theorem Optimality equation

$$V_{\beta}(i) = \sup_{a \in A(i)} \left\{ r(i, a) + \beta \sum_j P_{ij}(a) V_{\beta}(j) \right\}, i \in S$$

Conditions $|r(i, a)| \leq R_{\max} < \infty$ $\nLeftarrow \sum_j P_{ij}(a) V_{\beta}(j)$ is well defined

$$\Rightarrow (1-\beta) V_{\beta}(i) = \sup_a \left\{ (1-\beta) r(i, a) + \beta \sum_j (1-\beta) P_{ij}(a) V_{\beta}(j) \right\}$$

$$\begin{aligned} &= (V_{\beta}(i) - \beta V_{\beta}(i)) \\ &= \sup_a \left\{ r(i, a) + \beta \sum_j P_{ij}(a) V_{\beta}(j) - \beta V_{\beta}(i) \right\} \\ &= \sup_a \left\{ r(i, a) + \beta \sum_j P_{ij}(a) (V_{\beta}(j) - V_{\beta}(i)) \right\} \quad (*) \end{aligned}$$

Assumption $h(i) := \lim_{\beta \uparrow 1} [V_{\beta}(i) - V_{\beta}(a)]$ $\&$ $P(i) = \sup_a \left\{ \sum_j P_{ij}(a) P(j) \right\}$

So based on (*) we may want to write $P(i)$ as follows:

$$P(i) = \sup_a \left\{ r(i, a) + \sum_j P_{ij}(a) - \lim_{\beta \uparrow 1} [V_{\beta}(j) - V_{\beta}(a)] - [V_{\beta}(i) - V_{\beta}(a)] \right\}$$

$$= \sup_a \left\{ r(i, a) + \sum_j P_{ij}(a) [h(j) - h(i)] \right\}$$

$$\rightarrow \underline{P(i) + h(i) = \sup_a \left\{ r(i, a) + \sum_j P_{ij}(a) h(j) \right\}}$$

Part 2 why $P_{ii} \neq P$?

Condition $h_{\beta}(i) = V_{\beta}(i) - V_{\beta}(0)$ is uniformly bounded in i and β (Span bounded)
 $\exists M < \infty$ such that $\text{Span}(V_{\beta}) := \max_i V_{\beta}(i) - \min_i V_{\beta}(i) \leq M$

Theorem (Ross [1983, Thm 2.2]). There exists a sequence $\beta_n \uparrow 1$ and a scalar P and $h: S \rightarrow \mathbb{R}$ such that when $n \rightarrow \infty$,

(i) $(1 - \beta_n) V_{\beta_n}(0) \rightarrow P$ $\nrightarrow h_{\beta_n}(i) \rightarrow h(i) \quad \forall i \in S$

(ii) $P + h(i) = \sup_a \left\{ r(i, a) + \sum_j P_{ij}(a) h(j) \right\}$

If we choose $\alpha_{\beta}(i) \in \arg \max_{a \in A(i)} \{ \dots \}$ Then we have:

$$V_{\beta}(i) = r(i, \alpha_{\beta}(i)) + \beta \sum_j P_{ij}(\alpha_{\beta}(i)) V_{\beta}(j)$$

$$= r(i, \alpha_{\beta}(i)) + \beta \sum_j P_{ij}(\alpha_{\beta}(i)) (h_{\beta}(j) + V_{\beta}(0))$$

$$\rightarrow h_{\beta}(i) + V_{\beta}(0) = r(i, \alpha_{\beta}(i)) + \beta \sum_j P_{ij}(\alpha_{\beta}(i)) h_{\beta}(j) + \beta V_{\beta}(0)$$

$$\rightarrow h_{\beta}(i) + (1 - \beta) V_{\beta}(0) = r(i, \alpha_{\beta}(i)) + \beta \sum_j P_{ij}(\alpha_{\beta}(i)) h_{\beta}(j)$$

Based on (i) we have the following:

$$h(i) + P = r(i, \alpha^*(i)) + \sum_j P_{ij}(\alpha^*(i)) h(j)$$

Then we obtain the following:

$$P + h(i) = \sup_a \left\{ r(i, a) + \sum_j P_{ij}(a) h(j) \right\}$$

□

A Simple Counterexample!

$$S = \{1, 2\}, A(1) = A(2) = \{a\}$$

$$P_{11}(a) = 1, P_{22}(a) = 1$$

$$r(1, a) = r_1, r(2, a) = r_2 \Leftarrow r_1 \neq r_2$$

$$\Rightarrow P(i) = \lim_{\beta \uparrow 1} (1-\beta) V_{\beta}(i) = r_i \Rightarrow P(1) \neq P(2)$$

$\Leftarrow$ if $h_{\beta}(1) = 0$ then $h_{\beta}(2) = \frac{r_2 - r_1}{1 - \beta}$
then $\{h_{\beta}\}$ is not uniformly bounded here